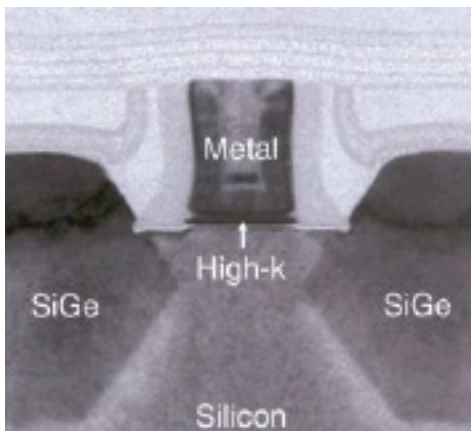


The incredible shrinking though, throws a wrench into the ideal. As transistors shrink, the wires connecting them shrink and the insulation that prevents the current to flow until the threshold is reached, gets worse. This can be likened to a leaking dam—the walls of which have grown holes letting the water through, when the dam shouldn't. As the pressure increases on the dam, the holes get larger, getting more water through. Thus, even when the dam should stop water (transistor OFF), it lets water through (leakage).

Shekhar Borkar, Intel Fellow and Director of Circuit Research puts it eloquently, "In a perfect switch, such as your light switch, when it is turned off, there is no current flowing through it, and the light bulb does not glow. If the switch is bad, then even though it is turned off there could be some residual current flowing through it, which may be so small that the light bulb does not seem to glow, but its there. Modern transistors are analogous to these bad light switches, they leak when they are turned off. And to make it worse, Moore's Law allows you to double these bad switches every two years, exponentially increasing the leakage every two years, and now it is becoming noticeable." A graph alongside this chilling declaration shows that at 45 nm (consumer processors have just reached this die size), the sub-threshold leakage current causes a power draw of 100 watts!

Thus far, you have seen die shrinks negatively effect the way the transistors work. The wires that connect



Intel's 45-nm processors are based on transistors with Hafnium-based high-k metal gate silicon technology

these nanomachines inside the processor together are not spared either. As the processor shrinks, the wires that connect the parts should shrink as well. The shrinking increases the resistance of these interconnects, which slows down the rate of current flow through them. So the interconnects get slower to transmit currents, but here's the catch—the transistors themselves are actually getting faster as they shrink. So you have one part of the processor—the brains really, running away, while the part that connects these speed demons is caught lag-

ging behind! This phenomenon is known as 'wire delay' and it is yet another reason why processor speeds can't just increase without solving some basic plumbing problems.

Taming The Beast

Taming first calls for plugging the plugs, or at least as many plugs as current technology can fix. The biggest hindrance to increasing clock speeds is leakage current. As we saw earlier, leakage without solutions could cause a processor to consume as much as 100 watts at 45 nm, and about 250 watts in total. One part of the solution is an implementation of a technology called Silicon on insulator (SOI), pioneered by IBM. Processors are generally built on a substrate of silicon. With SOI, the silicon substrate is replaced by a sandwich of silicon-insulator-silicon. The silicon junction is above an electrical insulator, typically silicon dioxide or sapphire. The insulating layer and topmost silicon layer also vary widely with application. SOI improves the insulation of the transistor gate and thus reduces leakage currents.

Another piece of the solution is use of so-called 'high-k' materials in building transistors. The 'k' here refers to the dielectric constant of a material. The dielectric constant tells you how effective a material is to storage of electrical charge, or the amount of electrical charge. For a transistor, k is a vital stat. The solution is replacing silicon dioxide, which has been used as a gate oxide material with a high-k material, such as Hafnium. As we learned earlier—with shrinking transistors, we get to a point where leakage currents due to tunnelling increase drastically, leading to increased power consumption and reduced device reliability. Replacing the silicon dioxide gate dielectric with a high-k material allows increased gate capacitance without the leakage effects.

Wire delay is reduced by using materials of low-resistance for the chip interconnects—generally copper, instead of aluminium. Another trick is to use more metal layers in the sandwich of the processor.

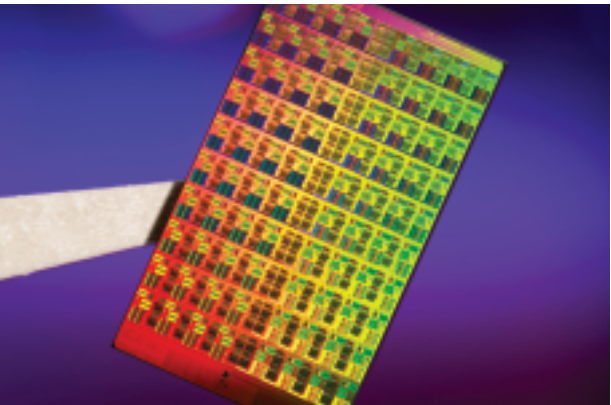
Increasing the metal layers helps pack the wires tighter together, which help keeps their length down, which in turn lowers their resistance and thus the delay. Further, the capacitance of the wires is also being reduced, this time by using low-k materials.

Engineers are also investigating the use of optical fibres instead of metal wires to carry information inside the chip. IBM, very recently, revealed a silicon photonic switch based on cascaded micro-ring resonators. This device can switch between nine channels with a bandwidth of about 40 Gbps per channel—allowing it to route data at speeds of 1 Tbps. It is capable of processing data with a switching speed of less than 2 ns. Fast enough to keep up with runaway processors? "It's certainly a big step in the right direction," said Yuri Vlasov, head of IBM's research into silicon nanophotonics. According to him, "It looks like our vision for on-chip optical interconnects is becoming more and more realistic." Sun Microsystems is also in the optical game, having recently won a contract from DARPA to use optical interconnect technology to stitch an upcoming supercomputer together. Sun's CTO and head of R&D, Greg Papadopoulos is optimistic "Optical communications could be a truly game-changing technology—an elegant way to continue impressive performance gains while completely changing the economics of large-scale silicon production." Optical interconnects are not just a solution to wire delay, but also a potential answer to the other problems arising from densely packing electrical and capacitive materials together, resulting in problems such as cross-talk, EMI and signal distortion.

The take-away from this discussion is that there isn't a single problem to tackle when it comes to engineering at nano scales. Similarly, there is no silver bullet to solve the myriad issues—it takes a mix of solutions to reduce the scale of issues faced, such that technology can keep trudging forward. But clearly, we are at a point where solutions need to catch up to the problems a bit more before we can hit that 10-GHz mark. Meanwhile, we need to redefine the playing field a bit.

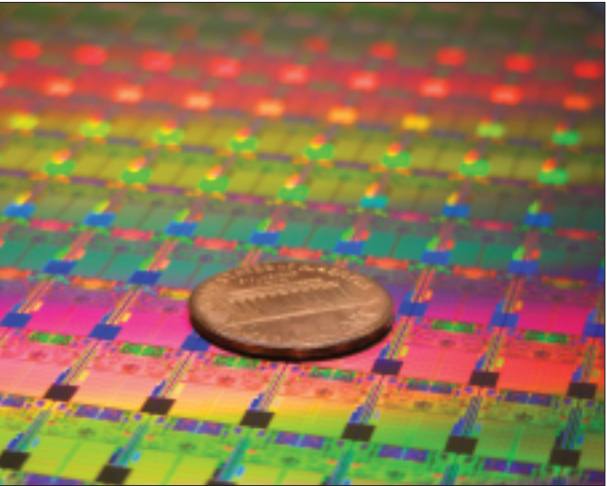
Our Very Own Hydra

Since processor speeds have reached something of a plateau, the path to progress lies in multiple cores. The idea is to combine several slower processors (slow in the relative sense here, they are still fast enough) to build a macro-processor. The trick then, is to split up work loads between these multiple constituent processors, such that, the total work done by the macro-processor is comparable to what could be done by a single super-fast processor. So goes the theory. And much like



The Teraflops Research Chip is Intel's first silicon tera-scale research prototype. This research project focuses on exploring new, energy-efficient designs for future multi-core chips, as well as approaches to interconnect and core-to-core communications.

the 10 GHz number was thrown by Intel in the time of the Netburst architecture, the multi-core era sees the concept of an 80-core chip, part of the so-called Tera-scale project inside the Intel labs. The theoretical applications of such many-core processor can be quite intriguing. One possibility, for example, is that of a data centre that is located entirely on a single microprocessor chip consisting of these 80-cores—each core connected via optical



Processors on an Intel 45-nm Hafnium-based High-k Metal Gate "Penryn" wafer photographed with a penny. These processors incorporate 410 million transistors for each dual-core chip, and 820 million for each quad-core chip.

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